

Computational Fluid Dynamics II

Lab 2 Part 1: Flow in Lid Driven Cavity

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Task 2.1

The entire 2.1 section describes how to preprocess, run and post-process cases involving isothermal, incompressible flow in 2D square domain.

Initially the flow is assumed to be laminar and we will solve it using **icoFoam solver** for laminar, isothermal, incompressible flow.

Later Reynolds number will be increased and the **pisoFoam or SIMPLEFoam solver** will be used for turbulent, isothermal, incompressible flow

Preprocessing

We have to set up cases by editing the case files. A case which is being simulated involves data for mesh, properties, geometry, control parameters(conditions) etc. Users should copy the cavity case directory to their run directory.

We can do this process by running these commands in the terminal.

```
cd $FOAM_RUN
```

```
cp -r $FOAM_TUTORIALS/incompressible/icoFoam/cavity/cavity .
```

```
cd cavity
```

cd means code directory

cp means copy

Mesh Generation

OpenFOAM can only run 3D simulations so, to run 2D simulations we consider a very small span length in z-direction (3rd direction) such that it works as 2D. For the cavity flow side length z-direction we take $d=0.1m$.

Once we are done with setting the dimensions of the body i.e., vertices, blocks etc. we also give types of boundary they are by giving types of walls. Then we mesh the entire body by using following command, **blockMesh**

If any error occurs it will give you an error message or else you are good to proceed further.

Viewing the Mesh

Before running the case it is better to view the mesh and check for any errors. To view mesh, we can use ParaView and it is conveniently launched on OpenFOAM, we can launch it in the background from the terminal using the command **paraFoam &** then the user have to select the "cavity.OpenFOAM" and click apply to view the mesh.

Boundary and initial conditions

Initial field setup will start at time $t=0$ sec, and this data is stored in a 0 sub directory of the cavity directory, this sub directory contains 2 files P(pressure) and U(velocity) whose initial values and boundary conditions must be set. File P contains 3 principal entries namely **dimensions**, **internalField** and **boundaryField**

Physical properties

Physical properties are stored in dictionary files in the "constant" directory. The main property to be specified for **icoFoam** is kinematic viscosity (ν), this can be done by editing "physicalProperties" dictionary. Initially we run the case with Reynolds number(Re) 10 and $Re=d|U|/\nu$.

Control

Input data relating to control of time, reading and writing of the solution data are in **controlDict** dictionary, this is a case control file and is located in the **system** directory. In this case of lid driven cavity we will start the run at time $t=0$ i.e., we set “startFrom” keyword to “startTime” and set it to 0.

For the end time we wish to reach a steady state where flow should be laminar, for this to happen the fluid or flow must pass through the domain at least 10 times. For the end time we wish to reach steady state and must specify the “stopAt” keyword as endTime and then set the endTime keyword to 0.5 (it should be 1.0 but we found that 0.5 is sufficient).

We need to set the time step, represented by the keyword “deltaT”. For temporal accuracy and numerical stability. When running icoFoam we need to specify a Courant number less than 1. $C_0 = \Delta t |U| / \Delta x$.

Discretization and linear-solver settings

User specifies the choice of FVD schemes in the “fvSchemes” dictionary in the system directory., for this case we only need pRefCell and pRefValue in the “PISO” sub-dictionary of the fvSolution dictionary. We will set both the values to 0. Changing any of these values will change the absolute pressure field, but not the relative pressure or velocity fields.

Running an application

In this case we run OpenFOAM using icoFoam solver by entering the command icoFoam in the case directory or else you can directly use the following command in the command prompt.

```
icoFoam -case $FOAM_RUN/cavity
```

The run is written in the terminal window. It tells the user the current time, maximum Courant number, initial and final residuals of all fields.

Now you can view the results in Paraview using the command paraFoam &

Post-processing

Once the simulation run is completed the user will be able to see time directories and can be viewed using paraFoam. We must first load the data at the required run time of 0.5 s. One should not run the case file when ParaView is opened because data will not be automatically loaded. In order to view the solution at $t = 0.5s$, the user can use the VCR Controls or Current Time Controls to change the current time to 0.5. These are located in the toolbars at the top of the ParaView window.

In the post processing we have many options to view our results (velocity and pressure fields). Such as: Colouring surfaces, cutting plane(slice), Contours, Vector plots, streamline plots, Increasing the mesh resolution.

We can also save those simulations as pictures or export them as animation which will save the files in the .png format at each time step.

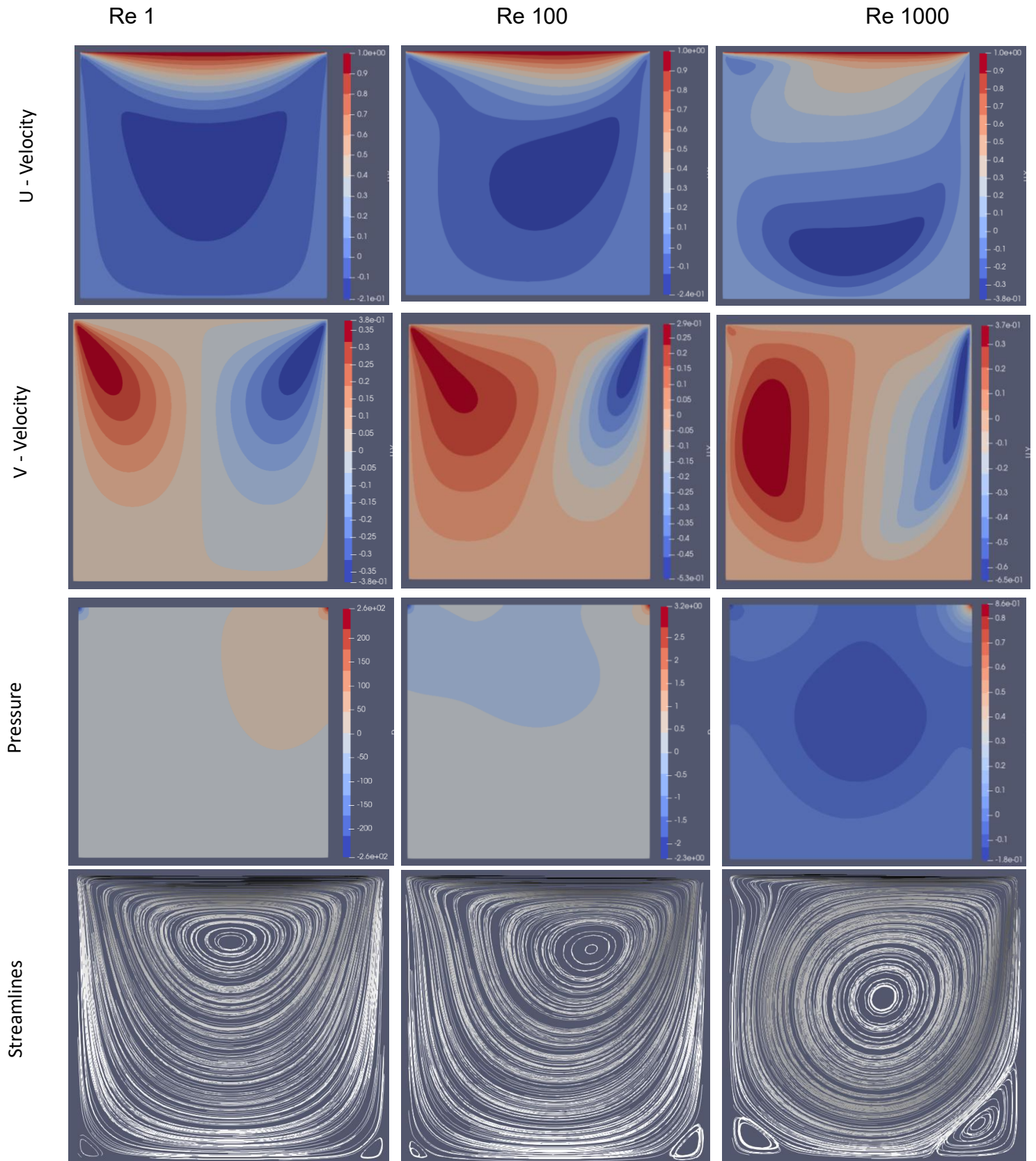
Further we can run the same case using SIMPLE Foam but we have to remove unnecessary files from the SIMPLE solver making it suitable for the lid driven cavity and also, we can control the mesh size and resolution along with parameters which need to be changed according to our setup/case.

Some times when we try to run the solver the system returns back with error message that the segmentation core got dumped, from my point of view it happens due two several reasons such as improper installation of Open FOAM, not updating the packages of that particular version, no proper establishment between Open FOAM and Paraview or not matching versions also if the user is using WSL

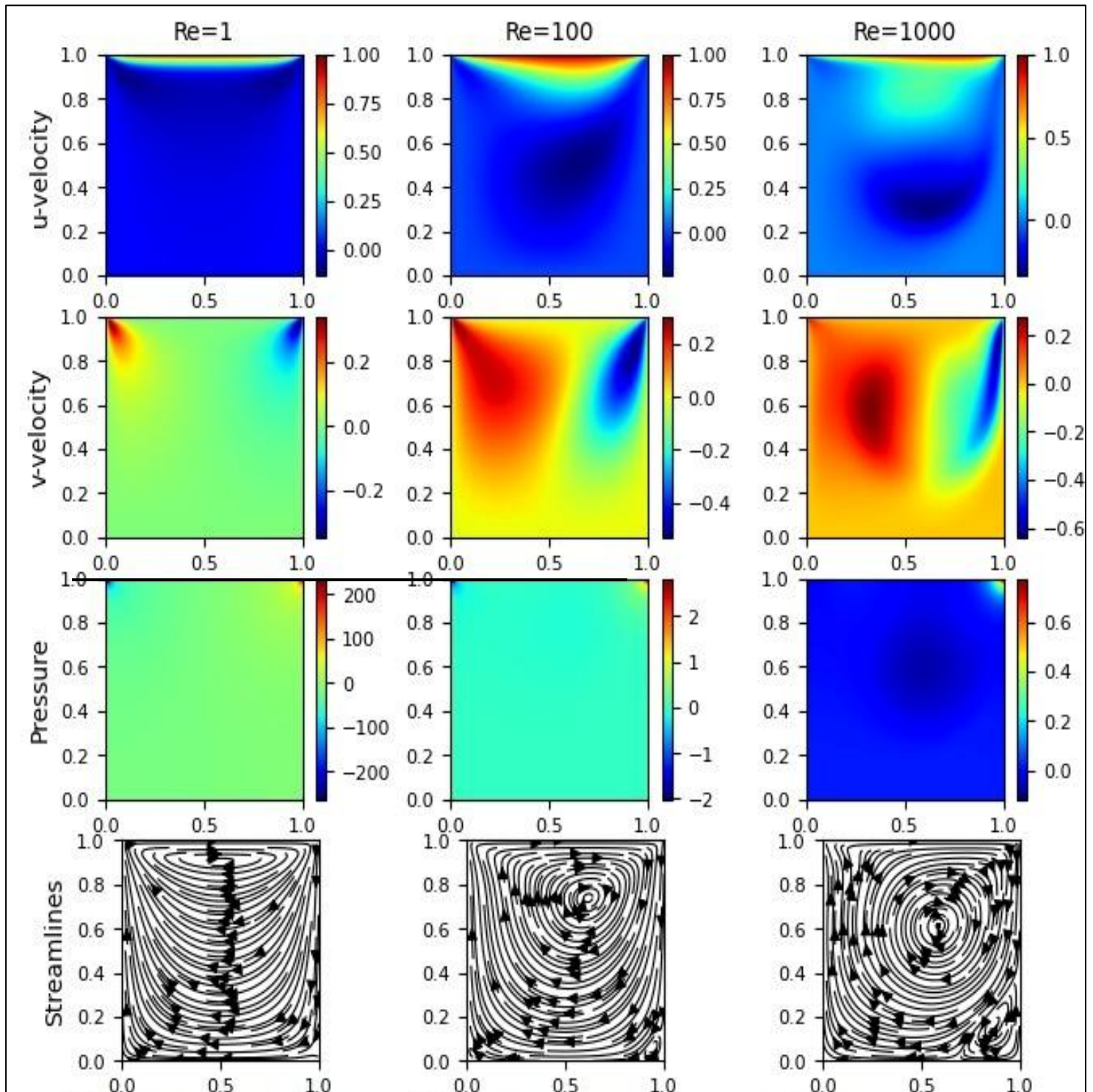
it is better to install Paraview for the windows directly from the website not to install it through WSL terminal. Every time if you are updating the mesh or increasing the size of the domain prior to running the solution user should run the `blockMesh` command, so the domain and the mesh gets updated to give the solution which belongs to that updated mesh and domain.

Task 2.2

Results obtained from OpenFoam



Results obtained from PYTHON

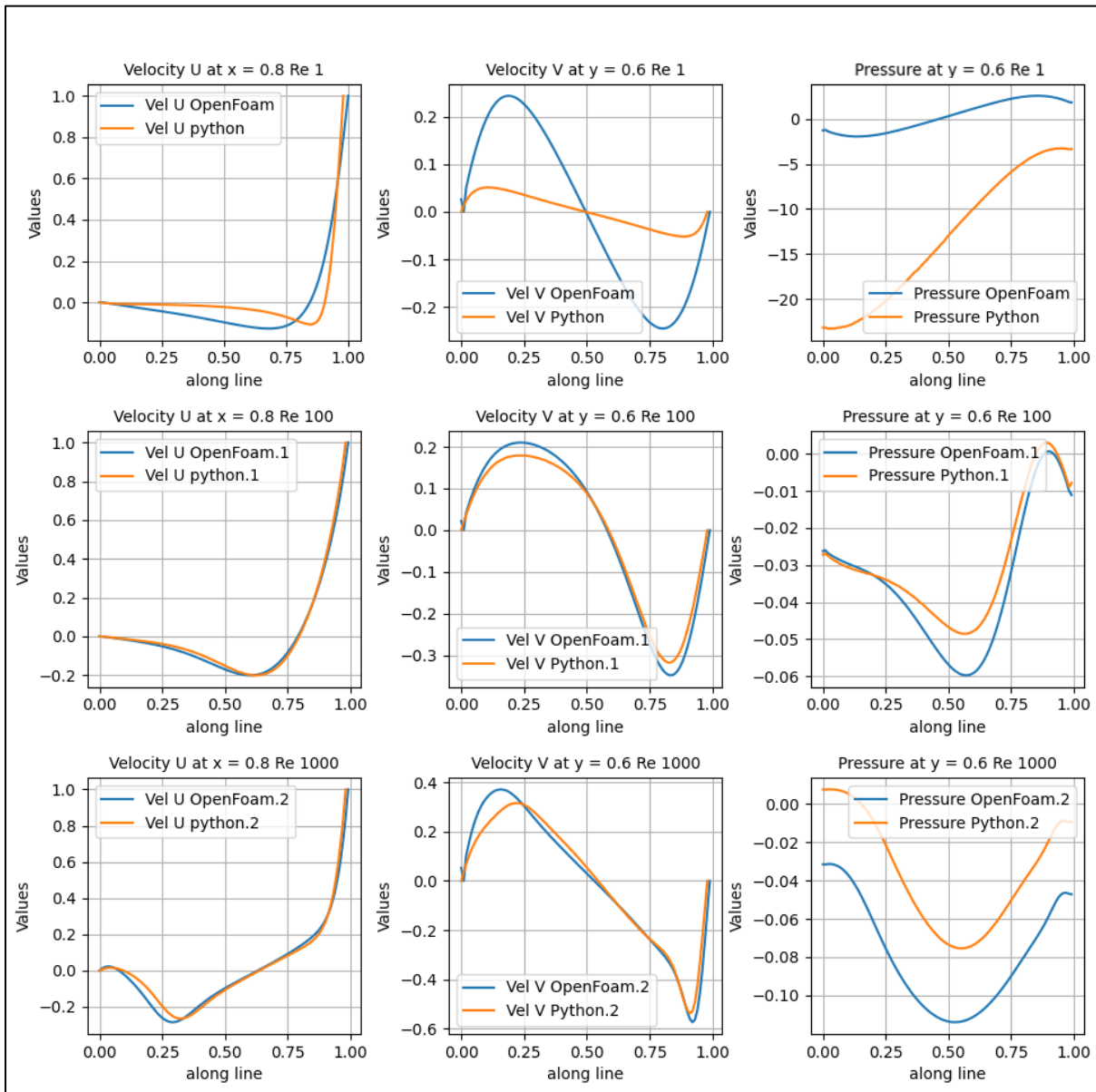


Steps to solve the laminar lid-driven cavity flow in OpenFOAM using the SIMPLE algorithm :

1. Copying the data from pitzDaily which is in simpleFoam by using the command "`cp -r $FOAM_TUTORIALS/incompressible/simpleFoam/pitzDaily .`" and copy the data to newly created folders namely re1, re100 and re1000.
2. The default settings for simpleFoam is "steady-state solver for incompressible, turbulent flow", but we need to solve laminar, so delete the other variables except U and p in directory 0.
3. Go to directory "system" in which we can edit the blockMeshDict, this can simply be done by copying the geometry from icoFoam/cavity which is provided by OpenFoam by default and make changes in the dimensions of the lid and no. of control volumes required(100 x 100).
4. Give the boundary conditions for U(no-slip condition) and p(neumann boundary conditions) in the directory 0.
5. Go to directory "constant" in which we can edit the momentumTransport by changing the simulationType from RAS to laminar and edit the other file physicalProperties by changing the "nu" with varying Reynolds number(re=1,100,1000) by the formula ($\nu = \frac{UL}{Re}$).
6. Go to directory "system" in which we can edit the file controlDict by changing the deltaT by calculating it with formula $Co = \frac{u \Delta t}{\Delta x}$, where Co is courant number(≤ 1), $u = 1$, $\Delta x = 1/100=0.01$ which in turn gives $\Delta t \leq 0.01$.
7. In the same directory, go to fvSolution where we can edit the solvers and algorithm by setting the SIMPLE algorithm, PCG solver for p and smoothSolver for U, set the tolerances for both U and p and then add relaxation factors also.
8. After doing and saving all these steps, generate the mesh by giving the command "blockMesh".
9. Run the simulation by giving the command "simpleFoam".
10. After the calculation are complete, give the command "paraFoam &" which will temporarily creates an OpenFoam file and views the results in Paraview.

Tasks 1.3

The locations chosen were $x = 0.8$ for U , $y = 0.6$ for V and P . These were such that maximum variation can be seen, especially near the vortexes. Good resemblance is seen for $Re\ 100$ and $Re\ 1000$. However, for $Re\ 1$, the python solver faced issues for convergence below $1e-4$. We can see that the U and V velocities show very similar trends near the vortexes. The plot of pressure for $Re\ 1000$ clearly shows how for SIMPLE algorithm, when p is the solution, $p+c$ is also a solution and thus, we see a clear shift of the same profile in the upward direction.



Contributions: All the members made themselves familiar with OpenFoam and ParaView

Hemanth Siva Kumar Ravanam – 1.1

Mihir Ajit Lonkar – 1.3

Mohan Kumar Meesala – 1.2